TRACE Fine-Tuning Framework for Robust Retrieval-Augmented Generation

Fine-tuning Method Based on Multi-agent Debate Traces

1 Multi-agent Debate to collect Traces

Expose correct/incorrect candidates and answer-shift trajectories under conflicts

Query Q

Who wrote The Mandalorian?

Retrieved Context C

Doc 1 (rel.)
Jon Favreau is an American filmmaker who created the Star Wars series The Mandalorian.

Doc 2 (misleading)
The Mandalorian was created by Dave Filoni

...

Response Model(weak)
(Target model to be fine-tuned)

Query-only response (parametric knowledge)
Dave Filoni

Critic Model (strong)
Check factuality, context reliability, and source selection.

Judge Model (medium)
Decide whether to stop and select key information.

Round 1 ... Round R

Response Model (weak)
Give initial answer and revise by feedback.

Debate Candidates *(examples)*

- **R1** Correct *Jon Favreau*
- **R2** Incorrect *Dave Filoni*
- **R3** Shifted *Jon Favreau is the director of the Mandalorian*

Final Debate Answer ***Dave Filoni***

Golden Answer ***Jon Favreau***

2 Construct Source-aware Training Records

Convert traces into compact records with source labels and functional

Source Label s_i *(knowledge source)*

internal	external	both
(parametric correct, context wrong)	*(parametric wrong, context correct)*	*(both sources support gold)*

Functional Type τ_i *(training role)*

Context Usage | Reliable Supervision | Context Resistance | Correct-to-Wrong Prevention | Dual-Source Hard-negative | Boundary Correction

One Debate Trace to Multiple Training Records

(The retrieved context is correct, so the s_i is external)

Candidate	Correct?	s_i	τ_i (Functional Type)
Query-only: Consor...	No	external	Context usage
Round 1: Consor...	No		Boundary correction
Round 2:Consort of the United Kingdom	Yes		Reliable supervision
Final:Consort of the United Kingdom	Yes		Context usage

3 Source-Aware Positive-Negative Objective

Learn which source to trust and generate

Source-Aware Positive-Negative Objective ($\mathcal{L}_{main}$)

Positive: Cross-Entropy (CE) for correct answers

$$\mathcal{L}_{CE}(x, y^+) = -\frac{1}{T^+}\sum_{t=1}^{T^+}\log p_\theta(y_t^+ \mid x, y_{<t}^+)$$

<answer> Jon Fav re au </answer> ...

Supervise the model to generate reliable answers.

Negative: Unlikelihood (UL) for incorrect answers

$$\mathcal{L}_{UL}(x, y^-) = -\frac{1}{\sum_t m_t + \epsilon}\sum_{t=1}^{T^-} m_t \log\left(1 - p_\theta(y_t^- \mid x, y_{<t}^-) + \epsilon\right)$$

<answer> Dave Filoni </answer>

Mask m_t=1 only for content tokens in the answer span.

Sample weight $\omega_i = w_{\tau(i)} \cdot \gamma_{s_i} \cdot r_i$

$w_{\tau(i)}$ *(by functional type)*: Reliable Supervision, Context Usage, Context Resistance, Dual-Source Hard-negative, Boundary Correction, Correct-to-Wrong Prevention (Low – High)

Preset weight for each training role.

γ_{s_i} *(by source label)*: internal, external, both

Balance signals across source conditions.

r_i *(stabilization for internal)*: History loss μ_{t-1}, Loss ℓ_t

$$r_i = \max(\beta, \mu_{t-1}(\ell_t + \epsilon))$$

Moderate reduces overly large penalties when inter is correct but external is wrong.

Answer Completeness Regularization Mechanism

1 Answer-tail Token Reinforcement

<answer> First Lady of the United States </answer>

a_t: 0.125 0.25 0.375 0.5 0.625 0.75

Gold Answer (inside answer span)
First Lady of the United States

earlier tokens (smaller weight)

later tokens (larger weight)

$$\mathcal{L}_{ans} = -\frac{1}{\sum_{t=1}^{T} m_t + \epsilon}\sum_{t=1}^{T} m_t(1 + \rho a_t)\log p_\theta(y_t \mid x, y_{<t})$$

Later tokens get larger weights to prevent truncation.

2 Premature Termination Suppression

First Lady of the United States </answer>

Too early!

Terminate here?

0.45 0.38 0.29 0.21 0.15 0.10 0.65

Inside answer span, reduce the probability of generating termination (eg. , EOS, or </answer>) too early.

$$\mathcal{L}_{end} = -\frac{1}{\sum_{t=1}^{T} m_t + \epsilon}\sum_{t=1}^{T} m_t \log(1 - p_t^{end})$$

Unlikelihood on termination tokens

Answer Completeness Loss

$$\mathcal{L}_{comp} = \mathcal{L}_{ans} + \lambda_{end}\mathcal{L}_{end}$$

Encourages complete well-bounded, and semantically complete answers.

Total Training Objective

$$\mathcal{L}_{total} = \mathcal{L}_{main} + \lambda_{comp}\mathcal{L}_{comp}$$

Correct candidate | Incorrect candidate | Judge decision | Debate rounds

Note: Debate is used only for training data construction. At inference time, the model answers in a single pass.